# Towards Optimized Hybrid Sparse Arrays for Wearable Ultrasonic Systems

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## Background, Motivation and Objective

In the recent years the usage of ultrasonic systems to measure vital parameters using wearable devices (e.g. smartwatches or patches) gained increasing interest. However, achieving sufficient *Signal-to-Noise-Ratio* (SNR) remains challenging due the small footprint of such systems. A solution to that problem might be *Hybrid Sparse Arrays* (HSA). Such arrays consist of strictly separated *transmit* (TX) and *receive* (RX) paths – thus giving potential for huge SNR improvements due to optimized hardware for their usage as TX/RX respectively and the usage of long coded signals. In this study, the design of HSAs, consisting of *Piezoelectric elements for TX and Capacitive micromachined ultrasonic transducer for RX* (PTCR), for the wearable use case are investigated.

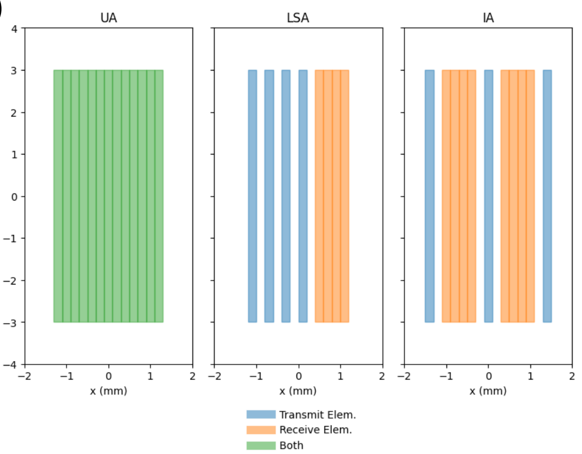
## Statement of Contribution/Methods

Several manufacturing-caused constraints must be considered while designing PTCR-HSAs, including that single channel CMUT are intractable to produce - thus only four channel chip configurations are considered in this study. Furthermore, sufficient spacing is required between transducer components and an overall footprint of $3\ \mathrm{mm} \times 6\ \mathrm{mm}$ should not be exceeded. Three array configurations are analyzed (see Fig. A): *Laterally Separated Array* (LSA), *Interleaved Array* (IA) and for comparison, a *Uniform Array* (UA). The arrays designs were emulated by selecting corresponding channels from a conventional probe (Esaote L533). *Contrast Ratios* (CR) of A-scans were reconstructed at the center of an anechoic inclusion with a $4\ \mathrm{mm}$ radius at a depth of $15\ \mathrm{mm}$, while the array center was laterally shifted to evaluate robustness to misalignment. The experiments were conducted using the ULA-OP 256 research ultrasound platform in combination with a Sun Nuclear Model 404 phantom.

## Results/Discussion

The results can be seen in Fig. B. As expected, the UA achieved the highest CR (max CR: $34.1\ \mathrm{dB}$) – however is not an HSA. Among the HSAs, the LSA (max CR: $24.6\ \mathrm{dB}$) consistently outperformed the IA (max CR: $16.2\ \mathrm{dB}$). The contrast degradation with increasing offset (UA: $\sim 3.5\ \mathrm{dB/mm}$, LSA: $\sim 2.5\ \mathrm{dB/mm}$, IA: $\sim 1.5\ \mathrm{dB/mm}$). The inferior contrast of IA could explain by the large transmit element pitch, which increases the prominence of grating lobes and degrades contrast. Overall, the results indicate that the LSA provide a more favorable compromise between hardware constraints and imaging performance for compact hybrid array designs.

A)




B)